Experiment 8

Linear Regression.

Program:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
df.info()
df.dropna(inplace=True)
df.info()
df.describe()
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,
test_size=0.2,random_state=23)
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
model.score(x_train,y_train)
model.score(x_test,y_test)
model.coef_
model.intercept_
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
print("Estimated Salary for {} years of experience is {}: "
.format(yr_of_exp,Salary))
```

Output

```
   YearsExperience  Salary
0              1.1   39343
1              1.3   46205
2              1.5   37731
3              2.0   43525
4              2.2   39891

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   YearsExperience 30 non-null     float64
 1   Salary          30 non-null     int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

Output

```
       YearsExperience         Salary
count       30.000000      30.000000
mean         5.313333   76003.000000
std          2.837888   27414.429785
min          1.100000   37731.000000
25%          3.200000   56720.750000
50%          4.700000   65237.000000
75%          7.700000  100544.750000
max         10.500000  122391.000000

LinearRegression()

0.960318254743891

0.918417084921423

array([[9281.30847068078]])

array([27166.73682891278])
```

Output

```
Enter Years of Experience: 44
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```